

Smart Power Bank Health Monitoring Using ESP32 and Machine Learning

ML MODEL USED: RANDOM FOREST



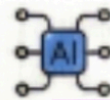
PROBLEM STATEMENT & OBJECTIVE

- Inaccurate SOC estimation in traditional battery systems
- Need for real-time monitoring of battery parameters
- Improve battery safety and performance
- Develop ESP32-based smart monitoring system
- Apply ML for SOC prediction and health analysis

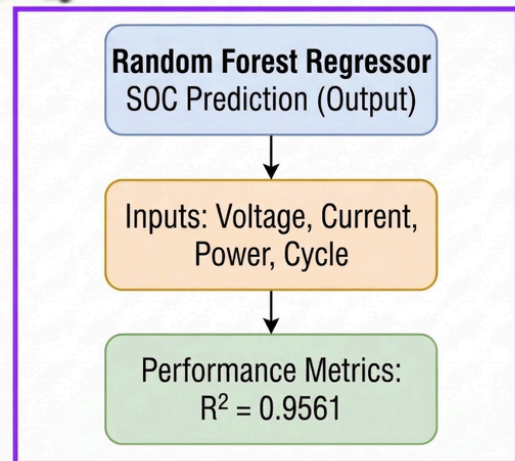


PROJECT FLOW

1. Sensor Data Collection (Voltage, Current, Temperature)
2. Data Transmission using ESP32
3. Data Processing in Python
4. SOC Prediction using Random Forest
5. Real-time Monitoring & Display



ML MODEL SETUP

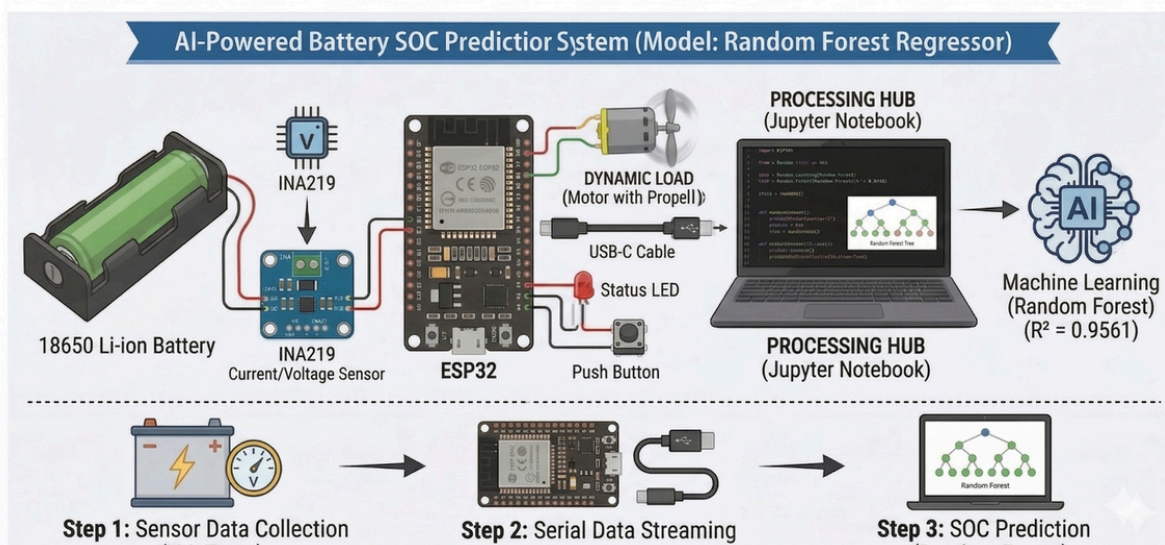


Random Forest Regressor

- Used for SOC prediction
- Trained on battery dataset
- Handles non-linear relationships
- Provides high accuracy ($R^2 = 0.9561$)



PROTOTYPE



ATHIKA MA , SATHISH M , AKILESHWARAN P , ASWATH P

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PROJECT OUTCOMES

The project demonstrates the application of embedded systems and Machine Learning to develop a smart battery monitoring system. It enables real-time tracking of battery parameters and accurate SOC prediction using a Random Forest model. The system improves problem-solving, analytical thinking, and practical implementation skills while ensuring safety through anomaly detection. The developed prototype is reliable, scalable, and suitable for real-world applications.



FUTURE SCOPE

- Mobile app for real-time monitoring
- Cloud-based data storage & access
- Advanced ML for better prediction
- Integration with EV systems
- Temperature-based safety monitoring
- Predictive maintenance

Scan this QR code to view the project demonstration video.



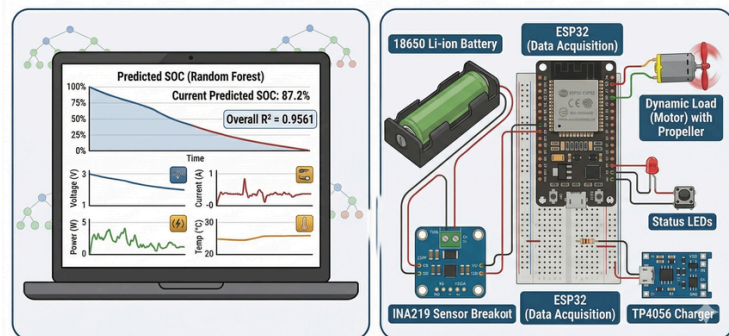
PERFORMANCE ANALYSIS

- Performance metrics
 - R^2 Score: 0.9561
 - Mean Absolute Error (MAE): $\pm 0.003\%$ SOC
 - Root Mean Square Error (RMSE): 0.000109



FINAL OUTPUT

FINAL OUTPUT



Smart Battery BMS: ML-Based SOC Prediction



TEAM DETAILS

(Enter Name & Roll No)

Roll No. - Name

23L211 - ATHIKA M A

23L261 - SATHISH M

24L431 - AKILESHWARAN P

24L432 - ASWATH P

The project currently uses the ESP32 as a data acquisition node to stream battery metrics to a PC where the Random Forest model runs; it is not yet a full Edge AI implementation